

# OXPLive MCP Server

Secure AI Access to the OXPLive Platform

|                |                                                   |
|----------------|---------------------------------------------------|
| Prepared for   | OrthoKinetix (ORX)                                |
| Prepared by    | Technijian                                        |
| Project        | OXPLive Modernization — Phase B Deliverable       |
| Date           | June 2026                                         |
| Status         | Pending Client Approval — Phase F (Build)         |
| Classification | Confidential — OrthoKinetix / Technijian Use Only |

|                   |               |           |                    |                  |
|-------------------|---------------|-----------|--------------------|------------------|
| 116               | 110           | 94        | 20                 | 100%             |
| Stored Procedures | API Endpoints | MCP Tools | Functional Domains | DB Call Coverage |

## 01 Executive Summary

---

OrthoKinetix operates the OXPLive practice management platform — a mature SQL Server system running the full DME order lifecycle: intake, benefits verification, insurance authorization, shipping, billing, re-billing, and collections across 13 workflow queues.

Technijian proposes to make that data safely available to AI assistants (Claude) through a purpose-built Model Context Protocol (MCP) server. The server connects Claude directly to the OXPLive database via a set of 116 curated stored procedures across 20 functional domains — without granting the AI any direct table access.

The governing rule: every AI request follows a single, auditable path — Claude calls a documented MCP tool → the tool executes an approved stored procedure → the stored procedure reads or writes the database. No exceptions. No inline SQL. No table-level permissions for the AI service account.

This document presents the complete scope of what will be built, the 116-procedure coverage matrix, the security and HIPAA controls in place, the delivery plan, and the approval request to begin Phase F (Build). The full engineering specification — including every stored procedure, its parameters, return schema, API endpoint, and MCP tool definition — is provided in the companion STP-API-MCP Coverage Matrix.

## 02 Why Now — The Current-State Gap

---

Technijian has been the active IT and development partner for OrthoKinetix since approximately 2015. As part of the ongoing OXPLive Modernization project (migrating from ColdFusion to .NET 8 / React 18 under Technijian SDLC v6.0), a complete database-call inventory was conducted in early 2026.

### What Was Found

The legacy OXPLive ColdFusion frontend makes hundreds of direct database calls across 13 workflow queues. These calls are not governed — they bypass stored procedures in many places, embed business logic in the UI layer, and contain hardcoded credentials. This creates three gaps that block safe AI access today:

- **No single governed access path.** The application reaches the database through a mix of stored procedures, inline SQL, and direct table reads. An AI assistant granted the same access would have uncontrolled write access to production data.
- **No audit trail by design.** Direct database calls leave no application-level record of who accessed what, when, or why — a HIPAA audit requirement.
- **No semantic boundary.** Without a procedure layer, there is no clean way to expose only what the AI should see while hiding raw PHI fields, financial data, and operational tables.

### The Solution

The OXPLive Modernization addresses all three gaps by establishing a complete, SP-Only database access pattern. Every frontend interaction — and every AI tool call — must go through a named, documented, version-controlled stored procedure. This is both a modernization best practice and the prerequisite for safe AI integration.

### 03 Architecture — DB-Direct Design

The OXPLive MCP server uses a DB-Direct architecture: Claude connects to the MCP server via Streamable HTTP, and the server executes stored procedures directly against SQL Server via a secure connection pool. There is no intermediate REST API layer for MCP access.

#### Request Path

|                                    |                                                                             |
|------------------------------------|-----------------------------------------------------------------------------|
| Claude (Desktop / Code / API)      | AI assistant — your staff or automated workflows                            |
| Streamable HTTP POST /mcp          | MCP 2025-03-26 spec · session-keyed · optional API-key                      |
| OXPLive MCP Server Node.js + mssql | 94 MCP tools · dispatches to named stored procedures                        |
| SQL Server — oxadmin schema        | Service account EXECUTE-only on oxadmin schema · no table permissions       |
| OXPLive Tables · Views · Functions | Production OrthoXpressDB — never directly accessible to the service account |

#### Why DB-Direct

The OXPLive Modernization Phase F will build a .NET 8 REST API, but the stored procedures are being designed now as part of Phase B. DB-Direct lets the MCP server go live the moment the stored procedures are tested — before the full API layer is complete — giving OrthoKinetix early AI access while the broader modernization continues.

Security posture is unchanged from an API-first design. The SQL Server service account (oxplive\_mcp\_svc) holds only EXECUTE on the oxadmin schema. It cannot read tables directly, run dynamic SQL, or access any object outside the curated procedure set.

#### Transport — Streamable HTTP

The server implements the MCP 2025-03-26 Streamable HTTP transport specification:

- **POST /mcp** — Creates or resumes a session; handles all JSON-RPC tool calls
- **GET /mcp** — SSE stream for server-initiated messages (session-keyed)
- **DELETE /mcp** — Gracefully closes a session
- **GET /health** — Returns { status: "ok", sessions: N }

## 04 Domain Coverage Matrix

The table below summarises the complete stored-procedure coverage across all 20 functional domains of the OXPLive system. Every domain corresponds to a distinct area of the OXPLive workflow UI. The full 116-row matrix — with individual SP parameters, return schemas, API endpoint definitions, and MCP tool specifications — is provided in the companion STP-API-MCP Coverage Matrix document.

| #  | Domain                             | STPs       | API Endpoints | MCP Tools | Business Purpose                                                                                   |
|----|------------------------------------|------------|---------------|-----------|----------------------------------------------------------------------------------------------------|
| 1  | <b>Auth &amp; Session</b>          | 5          | 4             | 2         | User authentication, token lifecycle, session management                                           |
| 2  | <b>Patient / Account</b>           | 18         | 16            | 16        | Core patient record — demographics, insurance, job creation, medical history, status, cancellation |
| 3  | <b>Insurance Authorization</b>     | 5          | 5             | 5         | Auth requests, line-item management, status tracking, approval workflow                            |
| 4  | <b>Queue &amp; Workflow</b>        | 7          | 7             | 6         | 13-stage workflow queue (Intake → Billing → Collections); job transitions, queue counts            |
| 5  | <b>Billing &amp; Claims</b>        | 5          | 5             | 5         | Fee schedules, billing search, claims processing, remittance management                            |
| 6  | <b>Scheduling</b>                  | 10         | 9             | 9         | Appointment creation, calendar search, therapist scheduling, 48-hour delivery alerts               |
| 7  | <b>Patient Experience</b>          | 7          | 6             | 5         | PSAT surveys, post-delivery call-back tracking, satisfaction scores, follow-up                     |
| 8  | <b>Collector Actions</b>           | 5          | 5             | 4         | Daily collector call logs, AR action tracking, daily-total summaries                               |
| 9  | <b>Products &amp; Inventory</b>    | 10         | 10            | 9         | Equipment catalog, SKU management, inventory levels, assignment to patients at delivery            |
| 10 | <b>Orders &amp; Purchasing</b>     | 3          | 3             | 3         | Purchase-order listing, vendor ordering, reorder tracking                                          |
| 11 | <b>Documents / Files</b>           | 4          | 4             | 3         | Document metadata, HIPAA-compliant signed download URLs for CMN, EDF, PHI files                    |
| 12 | <b>Directory, Users &amp; Reps</b> | 9          | 8             | 6         | Doctor/facility directory, sales rep management, referring physician lookup, staff directory       |
| 13 | <b>Time Records (HR)</b>           | 9          | 9             | 6         | Timecard entry, manager approval workflow, payroll-ready export                                    |
| 14 | <b>Lookups (Generic)</b>           | 4          | 4             | 2         | Shared reference data — insurance payer names, territories, status codes, dropdown values          |
| 15 | <b>Config</b>                      | 3          | 3             | 2         | System-wide configuration and company-settings retrieval                                           |
| 16 | <b>Reporting</b>                   | 4          | 4             | 4         | Billing collections, PSR, variance analysis, KPI dashboard data                                    |
| 17 | <b>Audit &amp; Activity Logs</b>   | 4          | 4             | 3         | HIPAA audit trail — user access log, data-change history, compliance events                        |
| 18 | <b>Compliance / Scheduled</b>      | 2          | 2             | 2         | Mass job-move operations and compliance batch processing                                           |
| 19 | <b>Yes/No Module (PA/CMN/EDF)</b>  | 1          | 1             | 1         | Three compliance gates: PA status, CMN receipt, EDF sign-off — block queue progression             |
| 20 | <b>Dashboard</b>                   | 1          | 1             | 1         | Real-time territory and user activity dashboard — session-aware, role-filtered                     |
|    | <b>TOTAL</b>                       | <b>116</b> | <b>110</b>    | <b>94</b> | <b>100% of OXPLive database calls covered</b>                                                      |

STPs = Stored Procedures. API Endpoints = planned .NET 8 REST endpoints (Phase F). MCP Tools = tools exposed to Claude via DB-Direct MCP server. Some STPs are INTERNAL and are not exposed as MCP tools.

## 05 Detailed Capability Reference

Every stored procedure in the OXPLive MCP server is listed below, grouped by domain. Each row shows whether the operation reads data or writes/updates it, the API endpoint it maps to in Phase F, and a plain-English description of what OrthoKinetix staff and AI assistants can do with it. In total: 73 read operations, 34 write operations, and 9 internal procedures not exposed to clients.

### Domain 1 — Auth & Session

| #   | Stored Procedure           | R/W  | API Endpoint | What It Does                                                                                       |
|-----|----------------------------|------|--------------|----------------------------------------------------------------------------------------------------|
| 1.1 | usp_Auth_EnsureUser        | INT  |              | INTERNAL — confirms/creates the Azure AD user on first login; called by API middleware only        |
| 1.2 | usp_Auth_GetCurrentUser    | READ | GET /api/me  | Returns the logged-in user's name, email, roles, division, and territory to populate the app shell |
| 1.3 | usp_Auth_GetIdpConfig      | INT  |              | INTERNAL — returns MSAL configuration for the SPA; agents use client_credentials instead           |
| 1.4 | usp_Auth_LogSessionHistory | INT  |              | INTERNAL — records every page navigation for the Admin activity log; called by request pipeline    |
| 1.5 | usp_Auth_RefreshToken      | INT  |              | INTERNAL — silently rotates access/refresh token pair when the SPA token expires; MSAL-managed     |

### Domain 2 — Patient / Account

| #    | Stored Procedure           | R/W   | API Endpoint                            | What It Does                                                                                          |
|------|----------------------------|-------|-----------------------------------------|-------------------------------------------------------------------------------------------------------|
| 2.1  | usp_Patient_Search         | READ  | GET /api/patients/search                | Find patients by name, account number, DOB, SSN-last-4, or phone                                      |
| 2.2  | usp_Patient_GetDetail      | READ  | GET /api/patients/{jobId}               | Retrieve complete patient record: demographics, insurance, medical info, referring doctor, status     |
| 2.3  | usp_Patient_Create         | WRITE | POST /api/patients                      | Create a new patient account with demographics, insurance, and medical intake data                    |
| 2.4  | usp_Patient_Update         | WRITE | PUT /api/patients/{jobId}               | Save edits to an existing patient record; changes are audited before/after                            |
| 2.5  | usp_Patient_GetSnapshot    | READ  | GET /api/patients/{jobId}/snapshot      | Get a complete printable snapshot: all products, billing codes, insurance, clinical notes in one call |
| 2.6  | usp_Patient_GetCoverage    | READ  | GET /api/patients/{jobId}/coverage      | Get payer name, policy number, group number, coverage type, and current auth numbers                  |
| 2.7  | usp_Patient_UpdateCoverage | WRITE | PUT /api/patients/{jobId}/coverage      | Update primary or secondary insurance coverage without touching demographics                          |
| 2.8  | usp_Patient_GetOrigRep     | READ  | GET /api/patients/{jobId}/orig-rep      | Get the original rep assigned at intake — preserves commission origination credit                     |
| 2.9  | usp_Patient_SetOrigRep     | WRITE | PUT /api/patients/{jobId}/orig-rep      | Set or update the originating rep when first assigned during intake                                   |
| 2.10 | usp_Patient_ListProducts   | READ  | GET /api/patients/{jobId}/products      | List all product/equipment line items with HCPCS codes, billing status, and auth notes                |
| 2.11 | usp_Patient_AddProduct     | WRITE | POST /api/patients/{jobId}/products     | Add a new product line item to the account; validates HCPCS code mapping                              |
| 2.12 | usp_Patient_UpdateProduct  | WRITE | PUT /api/patients/{jobId}/products/{id} | Update quantity or notes on a product line item                                                       |

|      |                             |       |                                                 |                                                                                               |
|------|-----------------------------|-------|-------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 2.13 | usp_Patient_RemoveProduct   | WRITE | DELETE<br>/api/patients/{fjobld}/products/{fid} | Remove a productline item; blocked if a claim has already been submitted                      |
| 2.14 | usp_Patient_ListNotes       | READ  | GET /api/patients/{fjobld}/notes                | Read all free-text notes on the account in chronological order                                |
| 2.15 | usp_Patient_AddNote         | WRITE | POST /api/patients/{fjobld}/notes               | Append a note to the account; notes are immutable once created                                |
| 2.16 | usp_Patient_GetAgreement    | READ  | GET /api/patients/{fjobld}/agreement            | Check whether the patient has accepted the service agreement and on what date                 |
| 2.17 | usp_Medical_UpdateDiagnosis | WRITE | PUT<br>/api/patients/{fjobld}/medical/diagnosis | Update ICD-10 diagnosis codes, referring doctor, body part, and procedure code                |
| 2.18 | usp_Patient_Cancel          | WRITE | POST /api/patients/{fjobld}/cancel              | Cancel the account with a required reason code and notes; writes a final status history entry |

## Domain 3 — Insurance Authorization

| #   | Stored Procedure       | R/W   | API Endpoint                                       | What It Does                                                                                       |
|-----|------------------------|-------|----------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 3.1 | usp_Auth_ListByPatient | READ  | GET /api/patients/{fjobld}/authorizations          | List all insurance authorizations: auth numbers, dates, payer, status, linked products             |
| 3.2 | usp_Auth_Create        | WRITE | POST<br>/api/patients/{fjobld}/authorizations      | Create a new insurance authorization record with auth number, dates, payer, and notes              |
| 3.3 | usp_Auth_Update        | WRITE | PUT /api/authorizations/{authId}                   | Update an existing authorization — correct dates, extend, or change status                         |
| 3.4 | usp_Auth_AddItem       | WRITE | POST /api/authorizations/{authId}/items            | Link a product line item to an authorization; required for billing to confirm each line is covered |
| 3.5 | usp_Auth_RemoveItem    | WRITE | DELETE<br>/api/authorizations/{authId}/items/{fid} | Remove a product from an authorization; blocked if the claim has been submitted                    |

## Domain 4 — Queue & Workflow

| #   | Stored Procedure           | R/W   | API Endpoint                              | What It Does                                                                                         |
|-----|----------------------------|-------|-------------------------------------------|------------------------------------------------------------------------------------------------------|
| 4.1 | usp_Queue_List             | READ  | GET /api/queues                           | List all 13 workflow queues visible to the current user with current job counts                      |
| 4.2 | usp_Queue_GetJobs          | READ  | GET /api/queues/{queueId}/jobs            | Get all patient accounts in a specific queue — paged, with name, status age, and rep                 |
| 4.3 | usp_Queue_TransitionJob    | WRITE | POST /api/queues/transition               | Move a patient account from one queue to another; validates the transition, writes status history    |
| 4.4 | usp_Queue_GetStatusHistory | READ  | GET /api/patients/{fjobld}/status-history | Get the complete queue-transition history for an account — every move with user and timestamp        |
| 4.5 | usp_Queue_GetActions       | READ  | GET /api/queues/{queueId}/actions         | List the permitted moves out of a queue — used to render the correct action buttons                  |
| 4.6 | usp_Queue_GetBillingStatus | READ  | GET /api/patients/{fjobld}/billing-status | Get billing status badge: Unbilled / Partially Billed / Fully Billed / Cash                          |
| 4.7 | usp_Queue_GetCounts        | READ  | GET /api/queues/counts                    | Get count of accounts in each of the 13 queues in one call — powers the sidebar badges and dashboard |

## Domain 5 — Billing & Claims

| #   | Stored Procedure                | R/W  | API Endpoint                            | What It Does                                                                                              |
|-----|---------------------------------|------|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| 5.1 | usp_Billing_ListEntities        | READ | GET /api/billing/entities               | List all billing entities (insurance companies, payers)                                                   |
| 5.2 | usp_Billing_Search              | READ | GET /api/billing/search                 | Search billable records by account, payer, or status — returns line items with charge and collection data |
| 5.3 | usp_Billing_GetFeeSchedules     | READ | GET /api/billing/fee-schedules          | Get contracted fee schedule amounts for a HCPCS code and billing entity                                   |
| 5.4 | usp_Billing_GetProductValuation | READ | GET /api/billing/valuations/{productId} | Get pricing and rental term options (capped rental, rent-to-own, purchase)                                |
| 5.5 | usp_Billing_GetReBillData       | READ | GET /api/patients/{jobId}/rebill        | Get billed line items flagged for re-billing due to denial, partial payment, or correction                |

## Domain 6 — Scheduling

| #    | Stored Procedure                | R/W   | API Endpoint                                     | What It Does                                                                                      |
|------|---------------------------------|-------|--------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 6.1  | usp_Schedule_GetByRange         | READ  | GET /api/schedules                               | List appointments filtered by date range, territory, or rep — powers the calendar view            |
| 6.2  | usp_Schedule_Create             | WRITE | POST /api/schedules                              | Create a new appointment for a patient — type, date/time, assigned rep, and notes                 |
| 6.3  | usp_Schedule_Update             | WRITE | PUT /api/schedules/{scheduleId}                  | Reschedule an appointment or update its type or notes                                             |
| 6.4  | usp_Schedule_Delete             | WRITE | DELETE /api/schedules/{scheduleId}               | Cancel and remove a scheduled appointment; also removes Exchange calendar sync                    |
| 6.5  | usp_Schedule_GetUnscheduled     | READ  | GET /api/schedules/unscheduled                   | Find accounts in the scheduling queue with no upcoming appointment                                |
| 6.6  | usp_Schedule_GetTypes           | READ  | GET /api/lookups/schedule-types                  | Get appointment type dropdown (delivery, follow-up, fitting, re-evaluation, etc.)                 |
| 6.7  | usp_Schedule_GetExchangeSync    | READ  | GET /api/schedules/{scheduleId}/exchange-sync    | Get the Outlook calendar sync status for an appointment                                           |
| 6.8  | usp_Schedule_SetExchangeSync    | WRITE | PUT /api/schedules/{scheduleId}/exchange-sync    | Link an appointment to an Outlook calendar event after pushing to Exchange                        |
| 6.9  | usp_Schedule_DeleteExchangeSync | WRITE | DELETE /api/schedules/{scheduleId}/exchange-sync | Remove the Exchange calendar link when an appointment is cancelled                                |
| 6.10 | usp_Scheduler_Get48HourAlerts   | READ  | GET /api/schedules/alerts                        | Get deliveries and follow-ups due within 48 hours for the user's territory — same-day alert panel |

## Domain 7 — Patient Experience (PSR)

| #   | Stored Procedure           | R/W  | API Endpoint                          | What It Does                                                                              |
|-----|----------------------------|------|---------------------------------------|-------------------------------------------------------------------------------------------|
| 7.1 | usp_PatientExperience_List | READ | GET /api/patients/{jobId}/experiences | List all patient contacts on the account — calls, inquiries, complaints — chronologically |



|     |                                         |       |                                               |                                                                                             |
|-----|-----------------------------------------|-------|-----------------------------------------------|---------------------------------------------------------------------------------------------|
| 7.2 | usp_PatientExperience_Create            | WRITE | POST /api/patients/{jobId}/experiences        | Log a patient contact: call attempt, inquiry, complaint, or scheduling contact with outcome |
| 7.3 | usp_PatientExperience_Update            | WRITE | PUT /api/patient-experiences/{peld}           | Update status, follow-up date, or notes on a patient contact record                         |
| 7.4 | usp_PatientExperience_GetTypes          | READ  | GET /api/lookups/patient-experience-types     | Get contact type dropdown (inquiry, scheduling, complaint, satisfaction, etc.)              |
| 7.5 | usp_PatientExperience_GetStatuses       | READ  | GET /api/lookups/patient-experience-statuses  | Get contact status dropdown (open, resolved, escalated, etc.)                               |
| 7.6 | usp_PatientExperience_GetComplaintTypes | READ  | GET /api/lookups/complaint-types              | Get complaint type dropdown — used when logging a formal patient complaint                  |
| 7.7 | usp_PatientExperience_GetTemplates      | READ  | GET /api/lookups/patient-experience-templates | Get message templates that pre-populate contact notes for common contact types              |

## Domain 8 — Collector Actions

| #   | Stored Procedure                   | R/W   | API Endpoint                              | What It Does                                                                                   |
|-----|------------------------------------|-------|-------------------------------------------|------------------------------------------------------------------------------------------------|
| 8.1 | usp_CollectorAction_List           | READ  | GET /api/collector-actions                | List collector actions for an account or by rep — calls made, voicemails, payment arrangements |
| 8.2 | usp_CollectorAction_Create         | WRITE | POST /api/collector-actions               | Log a new collection activity: call, voicemail, payment plan, referral to agency               |
| 8.3 | usp_CollectorAction_Update         | WRITE | PUT /api/collector-actions/{actionId}     | Update a collector action — revise follow-up date or add resolution notes                      |
| 8.4 | usp_CollectorAction_GetReasonCodes | READ  | GET /api/lookups/collector-action-reasons | Get reason code dropdown (called/no answer, payment plan set, referred to agency, etc.)        |
| 8.5 | usp_CollectorAction_GetDailyTotals | READ  | GET /api/collector-actions/totals         | Get aggregated collector performance stats by user and date range                              |

## Domain 9 — Products & Inventory

| #   | Stored Procedure             | R/W   | API Endpoint                                | What It Does                                                               |
|-----|------------------------------|-------|---------------------------------------------|----------------------------------------------------------------------------|
| 9.1 | usp_Product_List             | READ  | GET /api/products                           | Browse the product catalog filtered by category or active status           |
| 9.2 | usp_Product_Create           | WRITE | POST /api/products                          | Add a new product to the catalog with pricing and vendor info (admin only) |
| 9.3 | usp_Product_Update           | WRITE | PUT /api/products/{productId}               | Update product name, part number, pricing, or active status (admin only)   |
| 9.4 | usp_Product_GetBillingCodes  | READ  | GET /api/products/{productId}/billing-codes | Get HCPCS billing code mappings for a product — used in billing validation |
| 9.5 | usp_Product_GetBodyParts     | READ  | GET /api/lookups/body-parts                 | Get body part dropdown (Left Knee, Right Ankle, etc.) for medical intake   |
| 9.6 | usp_Product_GetCategories    | READ  | GET /api/lookups/product-categories         | Get product category dropdown (AFO, KAFO, TLSO, etc.)                      |
| 9.7 | usp_Product_GetManufacturers | READ  | GET /api/lookups/manufacturers              | Get vendor/manufacturer dropdown for product assignment                    |
| 9.8 | usp_Inventory_ListLocations  | READ  | GET /api/inventory/locations                | List warehouse/storage locations by site                                   |
| 9.9 | usp_Inventory_ListSites      | READ  | GET /api/inventory/sites                    | List inventory sites (warehouses and                                       |

|      |                      |              |                                            |                                                                                                                  |
|------|----------------------|--------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------|
|      |                      |              |                                            | offices)                                                                                                         |
| 9.10 | usp_Inventory_Assign | <b>WRITE</b> | PUT<br>/api/inventory/{equipmentId}/assign | Assign a specific piece of equipment to a patient — records serial number and date for stock-and-bill compliance |

## Domain 10 — Orders & Purchasing

| #    | Stored Procedure            | R/W         | API Endpoint                         | What It Does                                                                              |
|------|-----------------------------|-------------|--------------------------------------|-------------------------------------------------------------------------------------------|
| 10.1 | usp_Order_ListByPatient     | <b>READ</b> | GET /api/patients/{jobId}/orders     | List all orders and purchase orders on the account                                        |
| 10.2 | usp_Order_GetDetail         | <b>READ</b> | GET /api/orders/{orderId}            | Get full order detail including all line items with SKU, description, quantity, and price |
| 10.3 | usp_Purchasing_GetByPatient | <b>READ</b> | GET /api/patients/{jobId}/purchasing | Get sales order number, PO number, and invoice number for the account                     |

## Domain 11 — Documents / Files

| #    | Stored Procedure           | R/W          | API Endpoint                            | What It Does                                                                                        |
|------|----------------------------|--------------|-----------------------------------------|-----------------------------------------------------------------------------------------------------|
| 11.1 | usp_Document_ListByPatient | <b>READ</b>  | GET /api/patients/{jobId}/documents     | List all documents attached to the account: type, filename, upload date, uploader                   |
| 11.2 | usp_Document_GetMeta       | <b>INT</b>   |                                         | INTERNAL — retrieves blob path for signed URL generation; raw path never returned to client (HIPAA) |
| 11.3 | usp_Document_CreateRecord  | <b>WRITE</b> | POST /api/patients/{jobId}/documents    | Attach an uploaded file to the patient account; blob stored in Azure, metadata in database          |
| 11.4 | usp_Document_GetViewUrl    | <b>READ</b>  | GET<br>/api/documents/{documentId}/view | Get a 5-minute signed URL to view a patient document; access logged to HIPAA audit trail            |

## Domain 12 — Directory, Users & Reps

| #    | Stored Procedure               | R/W         | API Endpoint                           | What It Does                                                                   |
|------|--------------------------------|-------------|----------------------------------------|--------------------------------------------------------------------------------|
| 12.1 | usp_Directory_GetList          | <b>READ</b> | GET /api/directory                     | Browse the staff directory filtered by division or territory                   |
| 12.2 | usp_Directory_GetReps          | <b>READ</b> | GET /api/lookups/reps                  | Get the sales rep list, optionally filtered by territory                       |
| 12.3 | usp_Directory_GetTerritories   | <b>READ</b> | GET /api/lookups/territories           | Get territory lookup — used in scheduling, patient intake, and queue filtering |
| 12.4 | usp_Directory_GetDivisions     | <b>READ</b> | GET /api/lookups/divisions             | Get division/business-unit lookup                                              |
| 12.5 | usp_Directory_GetDoctors       | <b>READ</b> | GET /api/doctors                       | Browse the physician directory filtered by specialty                           |
| 12.6 | usp_Directory_GetPractitioners | <b>READ</b> | GET /api/practitioners                 | List clinical practitioners (PTs, OTs, RNs) for clinical assignment            |
| 12.7 | usp_User_ListAll               | <b>READ</b> | GET /api/admin/users                   | List all user accounts with roles, division, territory (admin only)            |
| 12.8 | usp_User_GetSecurity           | <b>READ</b> | GET /api/admin/users/{userId}/security | Get role assignments and territory access for a user (admin only)              |
| 12.9 | usp_User_GetRoleTypes          | <b>READ</b> | GET /api/admin/roles                   | Get all defined role types (Admin, Billing, Clinical, etc.)                    |

## Domain 13 — Time Records (HR)

| #    | Stored Procedure                  | R/W   | API Endpoint                             | What It Does                                                              |
|------|-----------------------------------|-------|------------------------------------------|---------------------------------------------------------------------------|
| 13.1 | usp_TimeRecord_List               | READ  | GET /api/time-records                    | List time record submissions by employee, pay period, or approval status  |
| 13.2 | usp_TimeRecord_GetDetail          | READ  | GET /api/time-records/{timeRecordId}     | Get a specific time record with all hourly or salaried line items         |
| 13.3 | usp_TimeRecord_Create             | WRITE | POST /api/time-records                   | Create a new time record header for a pay period                          |
| 13.4 | usp_TimeRecord_Update             | WRITE | PUT /api/time-records/{timeRecordId}     | Submit, approve, or reject a time record                                  |
| 13.5 | usp_TimeRecord_GetSalariedData    | READ  | GET /api/time-records/{id}/salaried-data | Get daily salaried entries and PTO allocations for a salaried time record |
| 13.6 | usp_TimeRecord_GetHourlyData      | READ  | GET /api/time-records/{id}/hourly-data   | Get hourly shift entries with start/end times and OT hours                |
| 13.7 | usp_TimeRecord_GetPTOTypes        | READ  | GET /api/lookups/pto-types               | Get PTO type dropdown (Vacation, Sick, Personal, Holiday, Bereavement)    |
| 13.8 | usp_TimeRecord_GetPaymentCalendar | READ  | GET /api/time-records/payment-calendar   | Get pay period dates for a year — valid week ranges for timecard entry    |
| 13.9 | usp_TimeRecord_GetHolidayCalendar | READ  | GET /api/time-records/holiday-calendar   | Get company holidays for a year — used in timecard calculations           |

## Domain 14 — Lookups (Generic)

| #    | Stored Procedure     | R/W   | API Endpoint                            | What It Does                                                                                                                 |
|------|----------------------|-------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| 14.1 | usp_Lookup_GetValues | READ  | GET /api/lookups/{tableName}            | Get dropdown values for any lookup table: cancellation reasons, document types, regions, territories, etc.                   |
| 14.2 | usp_Lookup_Create    | WRITE | POST /api/admin/lookups/{tableName}     | Add a new value to a lookup table (admin only); change is audit-logged                                                       |
| 14.3 | usp_Lookup_Update    | WRITE | PUT /api/admin/lookups/{tableName}/{id} | Update a lookup table value (admin only); change is audit-logged                                                             |
| 14.4 | usp_Lookup_Delete    | INT   |                                         | INTERNAL — soft-deletes a lookup value; admin UI action only, not an MCP tool (prevents agents from deleting reference data) |

## Domain 15 — Config

| #    | Stored Procedure      | R/W  | API Endpoint            | What It Does                                                                             |
|------|-----------------------|------|-------------------------|------------------------------------------------------------------------------------------|
| 15.1 | usp_Config_GetIdp     | INT  |                         | INTERNAL — MSAL configuration for the SPA; agents use client_credentials, not IDP config |
| 15.2 | usp_Config_GetAll     | READ | GET /api/config         | Get all system configuration values: session timeout, file size limits, feature flags    |
| 15.3 | usp_Config_GetCompany | READ | GET /api/config/company | Get company name, address, and branding for display in reports and print views           |

## Domain 16 — Reporting

| #    | Stored Procedure              | R/W  | API Endpoint                         | What It Does                    |
|------|-------------------------------|------|--------------------------------------|---------------------------------|
| 16.1 | usp_Report_BillingCollections | READ | GET /api/reports/billing-collections | Billing and collections report: |

|      |                                   |      |                                          |                                                                                                |
|------|-----------------------------------|------|------------------------------------------|------------------------------------------------------------------------------------------------|
|      |                                   |      |                                          | outstanding balances, payer performance, aging buckets                                         |
| 16.2 | usp_Report_DepartmentDeliverables | READ | GET /api/reports/department-deliverables | Department productivity: deliverables completed, time logged, productivity score by division   |
| 16.3 | usp_Report_PSR                    | READ | GET /api/reports/psr                     | Patient Satisfaction report: contact outcomes, complaint resolution, rep performance           |
| 16.4 | usp_Report_Variance               | READ | GET /api/reports/variance                | Pricing variance: billed amounts vs contracted rates — surfaces underbilling and margin issues |

## Domain 17 — Audit & Activity Logs

| #    | Stored Procedure        | R/W  | API Endpoint                 | What It Does                                                                               |
|------|-------------------------|------|------------------------------|--------------------------------------------------------------------------------------------|
| 17.1 | usp_AuditLog_Insert     | INT  |                              | INTERNAL — writes HIPAA audit events; called by API middleware only                        |
| 17.2 | usp_AuditLog_GetList    | READ | GET /api/admin/audit-logs    | Search the HIPAA audit log: PHI access, data changes, security events (admin/auditor only) |
| 17.3 | usp_ActivityLog_Insert  | INT  |                              | INTERNAL — records page views, logins, logouts; called by request pipeline only            |
| 17.4 | usp_ActivityLog_GetList | READ | GET /api/admin/activity-logs | Search user activity log: login, logout, and page-view events (admin only)                 |

## Domain 18 — Compliance / Scheduled

| #    | Stored Procedure              | R/W   | API Endpoint                         | What It Does                                                                              |
|------|-------------------------------|-------|--------------------------------------|-------------------------------------------------------------------------------------------|
| 18.1 | usp_Compliance_GetPendingJobs | READ  | GET /api/admin/compliance/pending    | List accounts awaiting compliance review that are overdue — daily compliance sweep        |
| 18.2 | usp_Compliance_MassMoveJobs   | WRITE | POST /api/admin/compliance/mass-move | Bulk-move accounts past the compliance hold period to the next queue; returns count moved |

## Domain 19 — Yes/No Module (PA/CMN/EDF)

| #    | Stored Procedure | R/W   | API Endpoint                    | What It Does                                                                                                                                   |
|------|------------------|-------|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 19.1 | usp_YesNo_Update | WRITE | PUT /api/patients/{jobId}/yesno | Update all three compliance gates at once: PA authorization, CMN receipt, EDF sign-off. PA Denied blocks the account from advancing to Billing |

## Domain 20 — Dashboard

| #    | Stored Procedure          | R/W  | API Endpoint                | What It Does                                                                                                                                                    |
|------|---------------------------|------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20.1 | usp_Dashboard_GetActivity | READ | GET /api/dashboard/activity | Get six headline KPIs: new intakes today, queue transitions, deliveries, documents uploaded, open collector follow-ups, overdue compliance — territory-filtered |

## 06 Security, HIPAA & Governance

OXPLive contains protected health information — patient demographics, insurance details, DME order data, and EDI 835/837 claims. Security controls are designed into every layer of this architecture, not added afterward.

| Control                          | Implementation                                                                                                                                                                                                                  |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Service Account Isolation</b> | oxplive_mcp_svc holds only GRANT EXECUTE ON SCHEMA::oxadmin. No table-level SELECT, INSERT, UPDATE, or DELETE. No dynamic SQL is possible.                                                                                      |
| <b>SP-Only Access Pattern</b>    | Every AI request resolves to a named, version-controlled stored procedure. Inline SQL from AI-generated queries is architecturally impossible.                                                                                  |
| <b>Audit Logging</b>             | Domain 17 exposes audit-log procedures. HIPAA-sensitive SPs (Document_GetViewUrl, all patient-record writers) trigger server-side audit entries recording user, action, timestamp, and affected record ID.                      |
| <b>PHI Field Masking</b>         | SSN and other PHI identifiers are masked at the stored-procedure level before they reach the MCP response payload. Patient-data fields require explicit role clearance.                                                         |
| <b>Document URLs</b>             | File access (usp_Document_GetViewUrl) returns HIPAA-compliant signed URLs with short-lived expiry. No raw file paths or permanent links are returned.                                                                           |
| <b>Transport Security</b>        | Streamable HTTP runs over HTTPS in production. Optional MCP_API_KEY header provides HTTP-level bearer auth. Network access restricted by firewall to authorised IP ranges.                                                      |
| <b>Session Isolation</b>         | Each Claude session receives a unique mcp-session-id. Sessions are server-memory-scoped; one session cannot access another session's state or results.                                                                          |
| <b>Compliance Domains</b>        | Domain 19 (Yes/No Module) enforces the three compliance gates — PA Status, CMN receipt, EDF sign-off — that must be satisfied before a job can progress. Domain 18 handles mass-move and scheduled compliance batch operations. |

This design supports OrthoKinetix's HIPAA compliance posture. It does not replace the organisation's existing policies, Business Associate Agreements, or staff training, which remain in place.

## 07 What Gets Built

---

### MCP Server — Node.js Application

The OXPLive MCP server is a Node.js / TypeScript application built on the official @modelcontextprotocol/sdk package. It uses the mssql library for SQL Server connectivity and Express for the Streamable HTTP transport.

- **DB Client** — Singleton mssql connection pool; executeSP(), querySP(), and querySPSingle() helpers that auto-prefix oxadmin. to every procedure name
- **20 Domain Tool Files** — One TypeScript file per domain, each exporting its MCP tool definitions and request handlers
- **Server Core** — createOxpliveMcpServer() registers all 94 tools and dispatches calls to the matching domain handler
- **Streamable HTTP Transport** — Express app with session map; supports multiple concurrent Claude sessions

### SQL Server Service Account

A dedicated, least-privilege SQL Server login is created specifically for the MCP server. No further permissions are required. The service account cannot read, write, or alter any table, view, or function directly.

Provisioning script: CREATE LOGIN oxplive\_mcp\_svc WITH PASSWORD = '<vault-managed>'; CREATE USER oxplive\_mcp\_svc FOR LOGIN oxplive\_mcp\_svc; GRANT EXECUTE ON SCHEMA::oxadmin TO oxplive\_mcp\_svc;

### Configuration & Deployment

All connection strings and secrets are managed via environment variables (never committed to source control). The server ships with a .env.example file and a config/claude-code.mcp.json snippet for immediate Claude Code integration.

### What Is NOT in Scope

- The .NET 8 REST API layer (Phase F — planned, not yet approved)
- An admin portal or management dashboard
- Changes to the existing OXPLive ColdFusion application
- Changes to the OrthoXpressDB schema or existing stored procedures

## 08 Delivery Plan

The MCP server delivery is structured in three phases. Each phase ends with a clear, demonstrable result that OrthoKinetix can test before the next phase begins.

| Phase   | Deliverable                 | Scope                                                                                                                                                                              | Exit Criteria                                                                                         |
|---------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Phase 1 | Stored Procedure Foundation | All 116 stored procedures authored, reviewed, and deployed to the OXPLive test database. Includes read-only procedures for all 20 domains plus write procedures for core entities. | All SPs pass unit tests against test DB; naming convention compliant; HIPAA-sensitive fields verified |
| Phase 2 | MCP Server — Build & Test   | Node.js MCP server with all 94 tools, DB-Direct wiring, Streamable HTTP transport, service account provisioning, and Claude Code integration.                                      | Claude can invoke all 94 tools against test database; audit log confirms all calls recorded           |
| Phase 3 | Production Deployment       | Production environment setup, credential vault integration, firewall rules, HTTPS, staff training session, and final sign-off from OrthoKinetix IT.                                | Server running in production; OrthoKinetix team can connect Claude and execute representative queries |

Detailed timeline and effort estimates are confirmed during the kick-off session and provided in a companion Statement of Work upon approval of this document.

## 09 What We Need From You

---

Approval of this document authorises Technijian to begin Phase 1. A short kick-off session is scheduled immediately afterward to confirm the items below. None of these decisions block approval — they shape the detailed plan.

- **Domain priority order.** Which of the 20 domains should be built first? Suggested default: Domains 1–5 — Auth, Patient, Insurance Auth, Queue, Billing.
- **Test database access.** Confirmation that Technijian may continue working against the existing test/UAT environment for Phase 1 and Phase 2.
- **Service account provisioning.** Authorisation to create the `oxplive_mcp_svc` login and grant it EXECUTE on the `oxadmin` schema.
- **Credential vault.** Preferred secret-management approach (Azure Key Vault, Windows Credential Manager, or Technijian-managed vault) for the service account password and API key.
- **Hosting location.** Where the MCP server will run — on-premises IIS, Windows service, Docker container, or Technijian-hosted.
- **Internal users.** Who at OrthoKinetix will use Claude with the MCP tools, and at what initial permission level (read-only vs. read/write access to write-capable domains).



## 10 Approval & Sign-Off

By signing below, OrthoKinetix authorises Technijian to proceed with the OXPLive MCP Server project as described in this document, beginning with Phase 1 (Stored Procedure Foundation). This approval covers the technical approach, domain scope, and phased delivery plan.

Specific timeline, effort hours, and fees are confirmed in a companion Statement of Work, which will be issued within five (5) business days of receiving this signed approval.

|                                                                                 |                                                                      |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------|
| <h2>OrthoKinetix</h2> <p>Authorised on behalf of OrthoKinetix / OrthoXpress</p> | <h2>Technijian</h2> <p>Prepared and submitted by Technijian Inc.</p> |
| <div> <div></div> <div></div> </div>                                            | <div> <div>Rohit Jain, Technijian</div> <div>June 2026</div> </div>  |
| <div> <div>Name (print)</div> <div>Title</div> </div>                           |                                                                      |

Questions? Contact Technijian at 949.379.8499 (main) or [rjain@technijian.com](mailto:rjain@technijian.com). The full engineering specification, STP-API-MCP Coverage Matrix, and Statement of Work are available on request.

## A Appendix — Full Domain Reference

### A.1 Domain Summary (authoritative counts)

| #  | Domain                     | STPs       | API Endpoints | MCP Tools |
|----|----------------------------|------------|---------------|-----------|
| 1  | Auth & Session             | 5          | 4             | 2         |
| 2  | Patient / Account          | 18         | 16            | 16        |
| 3  | Insurance Authorization    | 5          | 5             | 5         |
| 4  | Queue & Workflow           | 7          | 7             | 6         |
| 5  | Billing & Claims           | 5          | 5             | 5         |
| 6  | Scheduling                 | 10         | 9             | 9         |
| 7  | Patient Experience         | 7          | 6             | 5         |
| 8  | Collector Actions          | 5          | 5             | 4         |
| 9  | Products & Inventory       | 10         | 10            | 9         |
| 10 | Orders & Purchasing        | 3          | 3             | 3         |
| 11 | Documents / Files          | 4          | 4             | 3         |
| 12 | Directory, Users & Reps    | 9          | 8             | 6         |
| 13 | Time Records (HR)          | 9          | 9             | 6         |
| 14 | Lookups (Generic)          | 4          | 4             | 2         |
| 15 | Config                     | 3          | 3             | 2         |
| 16 | Reporting                  | 4          | 4             | 4         |
| 17 | Audit & Activity Logs      | 4          | 4             | 3         |
| 18 | Compliance / Scheduled     | 2          | 2             | 2         |
| 19 | Yes/No Module (PA/CMN/EDF) | 1          | 1             | 1         |
| 20 | Dashboard                  | 1          | 1             | 1         |
|    | <b>TOTAL</b>               | <b>116</b> | <b>110</b>    | <b>94</b> |

### A.2 The 13 Workflow Queues

The OXPLive platform routes DME orders through 13 named queues. Domain 4 (Queue & Workflow) provides the stored procedures that govern job transitions between queues.

| #  | Queue       | Purpose                                               |
|----|-------------|-------------------------------------------------------|
| 1  | Intake      | New DME orders received; initial eligibility check    |
| 2  | Benefits    | Insurance benefit verification in progress            |
| 3  | Auth        | Prior authorisation requested or under review         |
| 4  | Pending     | Awaiting patient or clinical information              |
| 5  | Shipping    | Order approved; equipment dispatch in progress        |
| 6  | Billing     | Delivered; claim submitted to payer                   |
| 7  | Re-Bill     | Claim denied or adjusted; re-submission in progress   |
| 8  | EDF         | Equipment Delivery Form outstanding — compliance hold |
| 9  | Collections | Overdue balance; active collector follow-up           |
| 10 | Yes/No      | PA Status / CMN / EDF compliance gates pending        |
| 11 | Inactive    | Order on hold; no active work                         |
| 12 | Cancelled   | Order cancelled; record retained for audit            |

|    |                   |                                            |
|----|-------------------|--------------------------------------------|
| 13 | Collector Actions | Collector daily-action tracking and totals |
|----|-------------------|--------------------------------------------|

### A.3 Companion Documents

- **STP-API-MCP Coverage Matrix** — All 116 stored procedures with full parameter lists, return schemas, API endpoint mappings, and MCP tool definitions. Provided as a separate A3-landscape PDF deliverable.
- **Statement of Work** — Issued within 5 business days of approval. Covers phased timeline, effort hours, and fees.